
Michael Murray

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📍 San Francisco, CA

Portfolio & More: mjm-murray.github.io

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SUMMARY

Mechanical engineer with 17 years of end-to-end product development experience across medical devices, utility-scale solar tracker hardware, industrial power, automation, and more – from concept and prototyping through production. Holds three U.S. patents in electromechanical design. Deep hands-on expertise in DFM for stamped, welded, machined, and molded parts; FEA; GD&T; fastener engineering; and rapid prototyping (CNC, 3D printing, molding), with a record of delivering compliant, cost-effective hardware under FDA, IEC 60601, ISO 13485, and other ANSI/ISO requirements. Leads cross-functional projects, qualifies suppliers, and mentors junior engineers.

PROFESSIONAL EXPERIENCE

Nevados Engineering

Oakland, CA

Senior Mechanical Engineer

April 2025 – October 2026

- Redesigned the stamped and welded tracker cradle assembly, improving structural performance and manufacturability; sized welds per AWS D1.1 and AISC 360.
- Performed linear and nonlinear FEA (SOLIDWORKS Simulation) on cradles, couplers, torque tubes, and bolted joints under wind downforce, uplift, and torsion to validate factors of safety.
- Led DFMEA of the coupler assembly and validated designs through stiffness, ultimate-strength, and tracker durability cycling tests.
- Owned fastener engineering and supplier quality: torque-tension and K-factor analysis (VDI 2230, ISO 16047), PPAP review, and supplier corrective actions; led coatings analysis and field-durability testing.
- Qualified gearmotor suppliers for Gen 1 and Gen 2 slew drives, including torque-curve evaluation, splined-shaft interface review, and an output-shaft failure investigation.
- Standardized SOLIDWORKS and PDM workflows, improving consistency and team efficiency.

Swope Design Solutions

San Francisco, CA

Senior Mechanical Engineer

April 2025 – Present

- Led design and build of diaphragm pump-to-catheter system for removing arterial calcified lesions, from proof of concept through tested prototypes.
- Created prototypes to establish proof of concept through CNC machining, molding, and 3D printing.
- Developed a positioning mechanism for a non-invasive surgical procedure to treat kidney stones.
- Built an automated data matrix scanning system, coding vision and analysis logic in C and Python.
- Designed a heater subsystem for a medical device, performing thermal analysis and benchtop testing; served as customer-facing project manager.
- Managed injection-molded part design and tooling, machined molds, and molded parts in silicone.
- Redesigned endoscopic robot components for injection molding, worked with overseas CM on moldability.
- Mentored junior engineers and ensured alignment with project milestones.

True Digital Surgery

Santa Barbara, CA

Senior Staff Mechanical Design Engineer

November 2020 – June 2022

- Designed mechanical subsystems of a positioning system for a robotic neurosurgical 3D microscope.
- Analyzed and prevented mechanical failures and confirmed factor of safety per IEC 60601 requirements.
- Performed thermal analysis to maintain mechanical integrity under heat application.
- Developed fit/function checks using GD&T in accordance with ASME Y14.5 standards.
- Co-authored QCIC system structure per ISO 13485 guidelines.

Surround Medical Systems

Morrisville, NC

Senior Mechanical Engineer

May 2019 – September 2020

- Led the mechanical department in designing a 3D x-ray positioning system from concept through production.
- Improved x-ray delivery to reduce patient radiation exposure while maintaining image quality.
- Co-authored a 510(k) submission for FDA clearance.
- Prevented mechanical failures through FEA and FMEA studies.

ABB**Senior Mechanical Engineer, R&D**

Raleigh, NC
March 2017 – May 2019

- Standardized transformer designs using common parts, reducing cost and lead times.
- Enhanced mechanical designs of transformers/enclosures to increase convection efficiency.
- Led a project to automate creation of drawings, models, and data, writing a standard for programming.
- Developed an outdoor-rated transformer cabinet incorporating weatherproof coating.

Ingersoll Rand**Design Engineer III**

Charlotte Area, NC
August 2014 – March 2017

- Designed parts/assemblies for complex, custom industrial centrifugal air compressors (2,000–7,000 cfm, 400–1,250 HP) to meet ANSI and ISO standards.
- Designed dual oil cooler system for 5000-cfm air compressors.
- Optimized airflow through pre-build assessments, improving compressor performance.

Newell Rubbermaid – Piedmont Hardware Brands**Mechanical Design Engineer**

Charlotte Area, NC
February 2013 – August 2014

- Designed high-volume sheet metal, die cast, injection-molded, and stamped parts.
- Utilized hand calcs and SOLIDWORKS Simulation for Finite Element Analysis, guiding new product designs.
- Prevented failure modes through Computer-Aided Engineering (CAE).

Sandvik**Manufacturing Engineer**

Mebane, NC
September 2011 – February 2013

- Generated CAD models, CNC programs through CAM, and drawings for high-volume metal-cutting tools.
- Reduced tooling costs by 30% via optimized CAM programs for 5-axis milling machines, lathes, and grinders.
- Designed and implemented a coolant reclamation system to optimize machining efficiency and cut waste.

ABB**Mechanical Design Engineer**

Lake Mary, FL
August 2009 – September 2011

- Invented a remotely operated device for industrial medium-voltage circuit breakers (5–15 kV); recognized with two patents (**U.S. Patents 9,876,335 and 8,654,513B2**).
- Developed a Modular Mechanism Operated Cell to retrofit circuit breakers (**Patent 8,878,088**).
- Improved circuit breaker thermal flow to prevent overheating and brazing issues.

TECHNICAL SKILLS

Skill	Tools / Scope	Level	Experience
DFM / DFA / DFx	Machined, stamped, welded cast, sheet metal, and molded parts	Advanced	16 yrs
CAD & Drafting	SOLIDWORKS, Creo, NX, Inventor, Onshape, Fusion	Advanced	17 yrs
FEA	SOLIDWORKS Simulation (static, nonlinear, modal, thermal)	Advanced	10 yrs
GD&T	ASME Y14.5 and more, tolerance stack analysis	Advanced	10 yrs
Product Development & PM	Jira, Confluence, Monday.com, and more	Advanced	11 yrs
Rapid Prototyping & 3D Printing	FDM, SLA, SLS, DMLS; CNC and manual machining; silicone molding	Advanced	11 yrs
Vendor Mgmt & Supplier Quality	PPAP and FAI, SCAR, fastener qualification, on-site supplier audits	Advanced	9 yrs
FMEA / DFMEA		Advanced	9 yrs
PDM / PLM	SOLIDWORKS PDM Professional Administration, Arena, Windchill	Advanced	9 yrs
Thermal Analysis	SOLIDWORKS Simulation, Flow Simulation	Intermediate	5 yrs
CAM	HSMWorks, NX	Intermediate	5 yrs
Regulations & Standards	FDA, IEC 60601, ISO 13485, ASME, ANSI, AWS D1.1, AISC 360, UL	Intermediate	5 yrs
Injection Molding & Plastics		Intermediate	6 yrs

EDUCATION**North Carolina State University, Mechanical Engineering, 2009**

Honors: Cum Laude, Dean's List, 2008 NCSU ME Design Award, Co-Op Program

ADDITIONAL NOTES

Portfolio, Manufacturing Process Experience, and more: <https://mjm-murray.github.io/>